THE UNITED REPUBLIC OF TANZANIA
ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS (OHONGSS)
KIGOMA REGION

FORM FOUR MOCK EXAMINATION

033/1 **BIOLOGY 1**

TIME: 3 Hours **YEAR: 2024**

INSTRUCTIONS

1. This paper consists of three sections A, B, and C with a total of eleven (11) questions.
2. Answer all questions in section A and B, and only two (2) questions in section C.
3. Section A carries sixteen (16) marks, Section B carries fifty four (54) marks and Section C carries thirty (30) marks.
4. All writings should be in blue or black ink except for drawings/diagrams which should be in pencil.
5. Cellular phone and any unauthorized materials are not allowed in the examination room.
6. Write your examination number on every page of your answer booklets provided.

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SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the following items (i)-(x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer sheet provided.

   i. Variation is possible in sexual reproduction because the process involves;
   - A. Plasmolysis
   - B. Hydrolysis
   - C. Mitosis
   - D. Ovulation
   - E. Meiosis

   ii. Straightening and bending of the arm involves contraction of the biceps and triceps muscles. Which of the following alternatives describes the state of the muscles when human arm is bent?
   - A. Triceps muscles contract while biceps relax
   - B. Biceps muscles relax while triceps contract
   - C. Both the biceps and triceps muscles relax
   - D. Biceps muscles contract while triceps relax
   - E. Both biceps and triceps muscles contract

   iii. Which of the following set of condition is necessary for seed germination
   - A. Temperature, soil and water
   - B. Water, temperature and food
   - C. Water, temperature and oxygen
   - D. Soil, oxygen and water
   - E. Water and carbon dioxide

   iv. The following balanced habit contains grasses, wild best, lions and bacteria, what would happen if lions were removed?
   - A. The number of bacteria would remain the same
   - B. The number of wild best would decrease
   - C. The amount of grasses would decrease
   - D. The amount of grasses would increase
   - E. The number of wild best would increase

   v. A certain grandfather visited a nearby health center where he saw a small building incinerator. What is the purpose of this:
   - A. For collecting liquid waste
   - B. For disposing plastics waste
   - C. For disposing gaseous waste
   - D. For burning hazardous waste
   - E. For collecting all types of waste

   vi. What will happen if phloem tissue is destroyed in green plants
   - A. Absorption of water in the plant body will stop
   - B. Absorption of minerals salts will stop
   - C. Transport of water and manufacture food will stop
   - D. Transport of manufactured food will stop
   - E. Transport of oxygen in the plant body will stop

vii. When the first day of menstruation was 8th April on which date is a women of 28 days will be of ovulation?

A. 26th March
B. 22nd April
C. 5th May
D. 16th April
E. 21st April

viii. A concave notch of kidney where renal artery, renal vein, ureter, nerve lymphatic vessel converge is called ;

A. Renal pelvis
B. Hilum
C. Hilium
D. Hillum
E. Hillium

ix. Kobelo started salivating as he saw a peeled half ripe mango. What part of the brain is concerned with Kobeles salvation

A. Hypothalamus
B. Medulla oblongata
C. Cerebellum
D. Cerebral hemisphere
E. Optical lobes

x. The gap at which the axon terminals of one neurone are in accossiation with another neurone or effector organ is called;

A. Inter neurone
B. Node of Ranvier
C. Dendrites
D. Synapse
E. Cell body

2. Match the phrases in list A with the responses in List B by writing the letter of the correct phrases from list B Besides the item number of list Ain the answer sheet provided

LIST A	LIST B
i. Growth which occurs when the rate of cell increase is higher than cell loss	A. Apical Meristems
ii. The type of cell division which result into two daughter cell	B. Lateral Meristems
iii. The kind of seed germination in which cotyledon emerge from the ground	C. Epigeal germination
iv. The ability of seed to germinate	D. Hypogeal germination
v. Cause the increase in length and heights of roots and roots	E. Moulting
vi. Shedding of exoskeleton in Arthropodss	F. Meiosis
	G. Mitosis
	H. Seed dormancy
	I. Seed viability
	J. Positive growth
	K. Negative growth

SECTION B (54 Marks)

Answer all questions from this section

3. (a) What would hypothetically likely to happen to an ecosystem, in case the following organisms are completely eliminated?
   (i) Green plant
   (ii) Fungi

(b) You have been given the following organism found at Mikumi game reserve. Study them and answer the questions below (Grass, Lizard, Grasshopper, Snake, Bacteria, Hawk and Bee)

i. Construct a food web
ii. How many food chain from the food web drawn above
iii. What will happen to an ecosystem above, If Lizard are eliminated?

4. (a)Given three organism P, Q and R; Organism P is found in aquatic environment, Q is found in terrestrial environment, and R is found in terrestrial environment but it has wings for flying. What kind of nitrogenous waste product is excreted by each of the above organism and why it is so?

   (b)Explain what will happen if the amount of sugar in the blood is not regulated?

5. (a)Evolution and its theories play a great role in biology studies, support the statement by any three points.

   (b)Briefly explain any observation and deduction made by Darwin in accomplishing his work (Give two observations and deductions)

   (c) State any two theories concerning with the origin of life.

6. (a) Various human activities tend to produce Carbon dioxide in the atmosphere, but the amount of carbon dioxide in the air tend to remain less or more constant. Explain how is this possible?

   (b) Why breast feeding is more important than bottle feeding?

7. If a pancreas as endocrine gland was by mistakenly removed during surgery from a patient; describe six (06) problems which the patient would get.

8. (a) A black rabbit homozygous for coat colour was crossed with a white rabbit of unknown genotype. The offspring was all black. By means of diagram and symbols find the genotype of parents and offspring.

   (b)What is sex linked characteristics?

   (c) If coat colour in rabbit in question 8(a) is a sex linked characteristic, what would be the genotypes and phenotypes offspring produced from a cross between black male and white female?

SECTION C (30 Marks)

Answer only two (2) questions in this section

9. Mwaitege lives in Mbeya city. During holidays he travelled to his uncle who lives in Dar es Salaam. When he came near to Dar es Salaam he claimed the highest degree of temperature within his body. As biology student, explain five processes that would occur to the Mwaitege's body to adopt the hot condition of Dar es Salaam.

10. Irresponsible sexual behaviour among youths poses a problem to the Tanzanian community. Justify this statement by elaborating six effects.

11. Most of Tanzanian societies in rural areas where there is no electricity face the problem of food spoilage because they do not have the knowledge of preventing their food from going bad. As a form four student, explain what you will advise them to solve this problem.(Any six (6) points)